Large Data Set: AQA

Questions 1–12

Question 1. State the following facts about the Large Data Set:

a) The five makes of vehicle.
b) The years in which the vehicles were registered.
c) The regions in which the registered keepers live.
d) The years in which the vehicles were manufactured.
e) The 5 types of propulsion.
f) The 9 types of body type.

Question 2. State the units of the following:

a) CO_2 (carbon dioxide).
b) CO (carbon monoxide).
c) NOX (oxides of nitrogen).
d) part (particulate emissions).
e) hc (hydrocarbon emissions).
f) mass.
g) engine size.

Question 3. Explain why, when sampling from the Large Data Set, that blanks may be found in some the data.

Question 4. Kerry samples a car from the Large Data Set and notes down that its mass, particulate emissions, and engine size are all 0.
In each case, state whether Kerry may have made a mistake or has correctly noted the details of the car.

Question 5. For each of the following, given a sample is being taken from the Large Data Set, write down the name of the sampling method used.

a) The order of the cars is randomised, then every 100th is noted.
b) The order of the cars is randomised, then the first 10 of each type of make is noted to obtain 50 cars.
c) The data set is sorted by makes and the proportions of each type calculated. Using a random number generator and the reference number, each make is selected to the same proportion as the full data set.

Question 6. Martha uses the complete Large Data Set and finds that the mean mass of a car in 2002 is 1278.7 kg and in 2016 is 1416.6 kg. She finds out that the average mass of a human in 2002 and 2016 was 75.9 kg and 78.0 kg respectively.

a) She says the actual mass of the cars is 1202.8 kg and 1338.6 kg. Explain why Martha is incorrect.
b) She claims that all vehicles in the UK are heavier in 2016 than in 2002. Comment on the validity of Martha's claim.

Question 7. The table below shows a sample of 10 vehicles from the Large Data Set.

Make	Region	Engine Size
Vauxhall	London	1398
Volkswagen	London	1198
Ford	North West	999
Volkswagen	London	1598
Vauxhall	South West	1398
Ford	London	998
Toyota	London	1398
Vauxhall	South West	1398
BMW	London	2171
Ford	North West	1498

a) Find the mean and the standard deviation of the engine size of this sample.

b) Give a reason why the sample used above may have biased data.

Any value more than 2 standard deviations from the mean can be identified as an outlier.

c) Find any outliers in this sample of engine size.

The mean engine size for the whole of the Large Data Set is 1646.2 cm^3 and the standard deviation is 456.0 cm^3

d) Give one comparison of central tendency for this sample vs the full data set.

e) Give one comparison of spread of this sample vs the full data set.

Question 8. The table below shows the CO_2 emissions (g/km) in 2002 and 2016, for all the diesel cars in the Large Data Set.

	2002	2016
$\sum x$	46 335	127 096
Number of cars	302	1 097

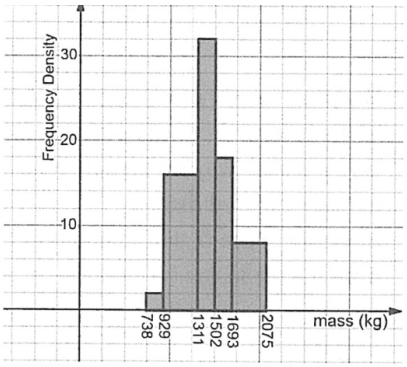
a) Find the mean of the CO_2 emissions for 2002 and for 2016

The mean CO_2 emissions for all the cars registered in 2016 in the Large Data Set is 120.37. There are 2529 total cars in the Large Data Set in the year 2016.

b) Find the mean of all the CO_2 emissions for all the vehicles in 2016 which are not diesel.

c) It is claimed that the increase in CO_2 emissions is caused by the electric cars which feature solely in the 2016 data. Explain, with reference to the Large Data Set, why this cannot be the case.

Question 9. A set of cars were randomly sampled from the Large Data Set and the masses have been displayed in the histogram below.



It is known that the bar between 1693 kg and 2075 kg represents 8 cars.

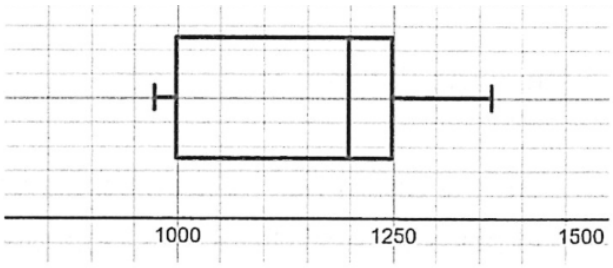
a) Calculate how many cars have been sampled altogether.
b) Estimate the mean mass of a car in this sample.
c) Find the probability of picking a car of mass between 1120 kg and 1693 kg.

Question 10. The Keeper Title of a sample of 250 cars are taken from the Large Data Set and a probability distribution is given:

Keeper Title	Probability
Male	$\frac{2}{5}$
Female	k
Unknown	$\frac{4}{125}$
Company	$\frac{9}{25}$

a) Find the value of k
b) State the number of each of the Keeper Title sampled.
c) Give one reason why the sample is representative of the Large Data Set.

Question 11. A set of 100 cars were sampled from the Large Data Set and the engine sizes of these cars are displayed in the boxplot below:



Using your knowledge of the Large Data Set, explain why it is possible that it was not sampled randomly.

Question 12. Prerequisite knowledge for Q12: Normal Distribution The CO_2 emissions, x g/km, from a sample of 40 cars are taken from the Large Data Set.
The summary statistics are as follows:

$$\sum x = 5592, \qquad \sum x^2 = 802\,846$$

a) Calculate the mean and the standard deviation, where $s_x = \sqrt{\dfrac{\sum x^2 - \frac{(\sum x)^2}{n}}{n-1}}$, giving your answer to 4 significant figures.

This data is believed to be normally distributed. Find the probability that the CO_2 emissions are

b) greater than 150 g/km.

c) exactly 114 g/km.